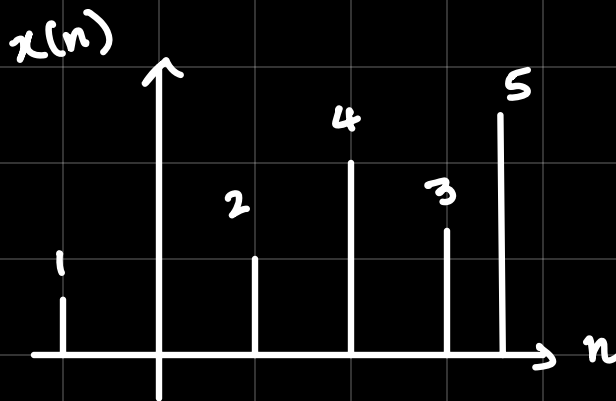


**Discrete Time Signal:** if a signal is defined at a discrete instant of independent time such a signal is called DTS.

Time =  $n$       Signal =  $x[n]$  /  $x(n)$



$$x(n) = \{1, 0, 2, 4, 3, 5\}$$

$\begin{matrix} -1 & 0 & 1 & 2 & 3 & 4 \\ \uparrow & & & & & \end{matrix}$

**Representation of DTS:**

1) Functional

$$x[n] = \begin{cases} 1 & n=-2 \\ 2 & n=-1 \\ 3 & n=0 \\ 4 & n=1 \\ 5 & n=2 \\ 6 & n=3 \end{cases}$$

2) Tabular

| $n$    | -2 | -1 | 0 | 1 | 2 | 3 |
|--------|----|----|---|---|---|---|
| $x[n]$ | 1  | 2  | 3 | 4 | 5 | 6 |

3) Sequence

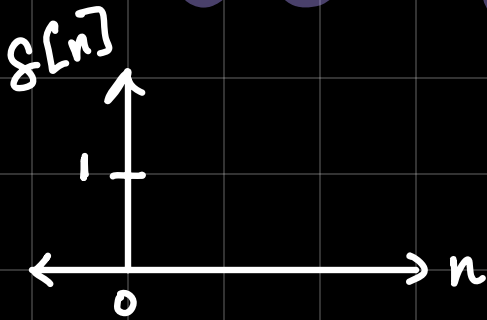
$$x[n] = \{1, 2, 3, 4, 5, 6\}$$

$\begin{matrix} \uparrow \\ 3 \end{matrix}$

# Standard DTS:

## ① Impulse

$$f[n] = \begin{cases} 1, & n=0 \\ 0, & n \neq 0 \end{cases}$$



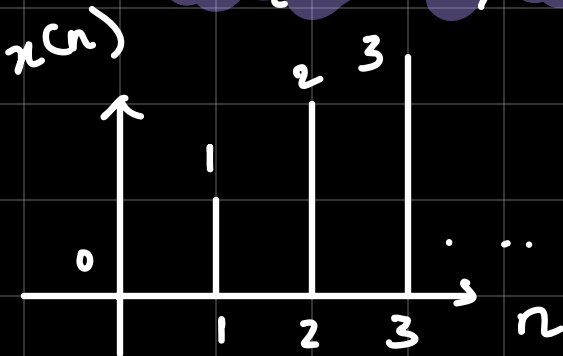
## ② Unit Step

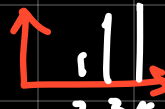
$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



## ③ Ramp

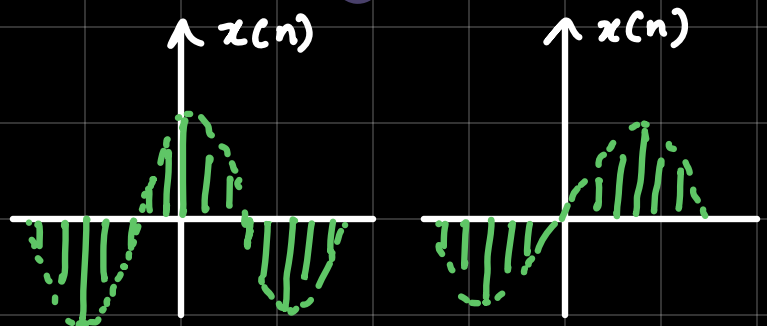
$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



\* shifted ramp  $\Rightarrow$  

## ④ Sinusoidal

$$x[n] = A (\cos \omega_0 n + \theta) = A (\sin \omega_0 n + \theta)$$



$$x[n] = x_a(t) \big|_{t=nT}$$

↙ sampling

$$T = 1/f_s \text{ or } f_s = 1/T$$

## ⑤ Exponential - Decay

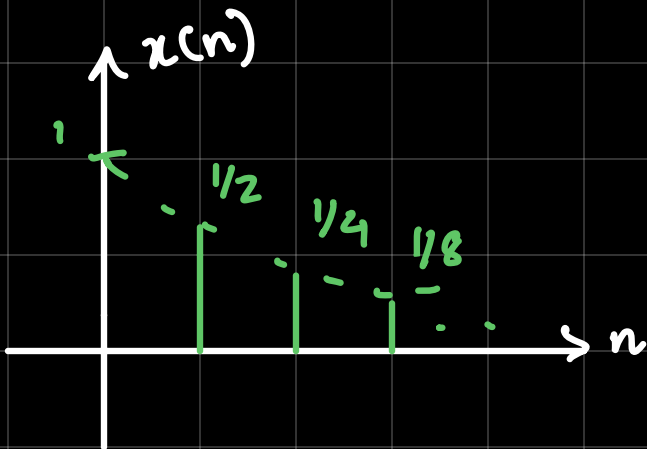
Growth

$$x[n] = a^n ; n \geq 0$$

$$= 0 ; n < 0$$

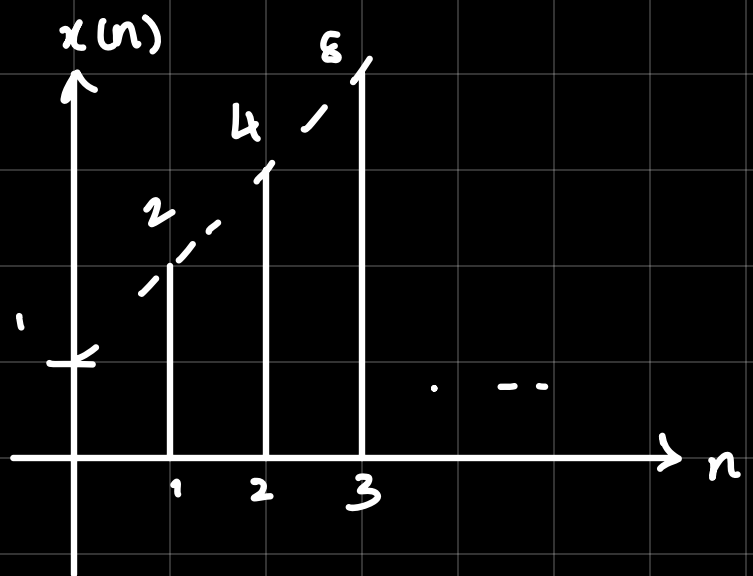
let  $0 < a < 1 \Rightarrow a = 1/2$

$$x[n] = (1/2)^n$$



$$a = 2$$

$$x[n] = (2)^n$$



## Basic Operations

-  $\rightarrow$  RIGHT

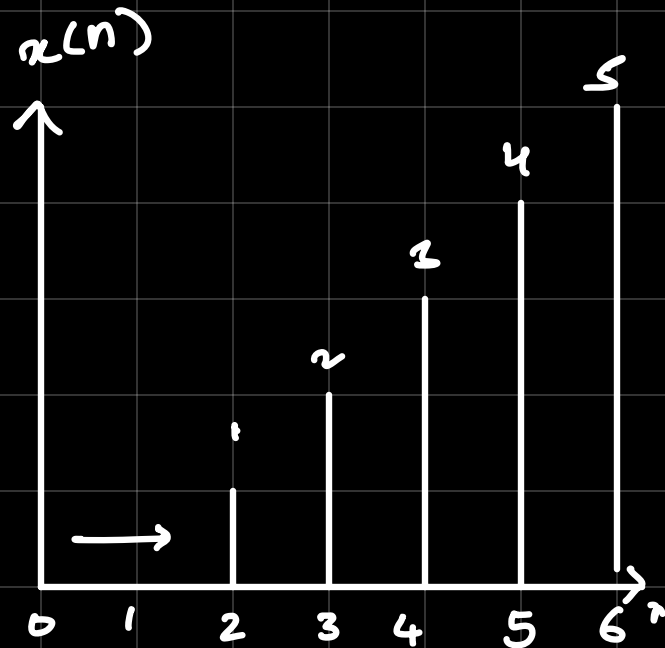
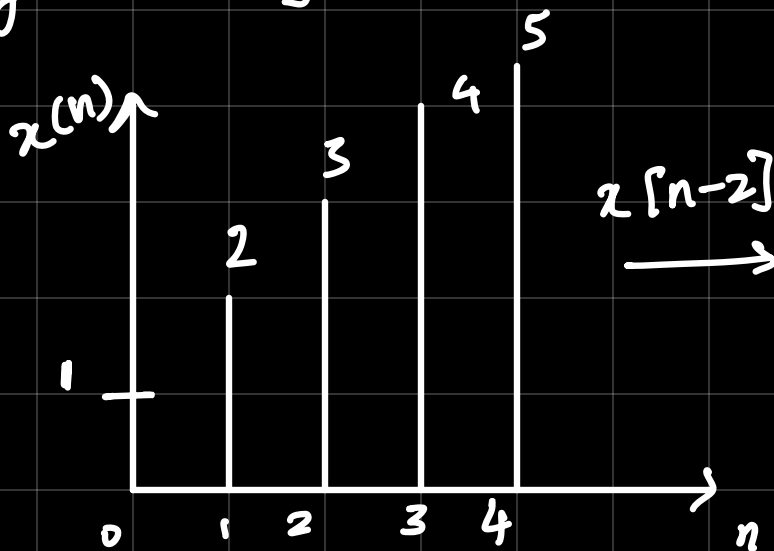
+  $\rightarrow$  LEFT

### ① Time Delay / Time Shifting

$x[n-k]$

$\rightarrow$  delay by  $k$  units  $\rightarrow$  Shift Right

eg:  $x[n-2]$



$$x[n-2] = \{0, 0, 1, 2, 3, 4, 5\}$$

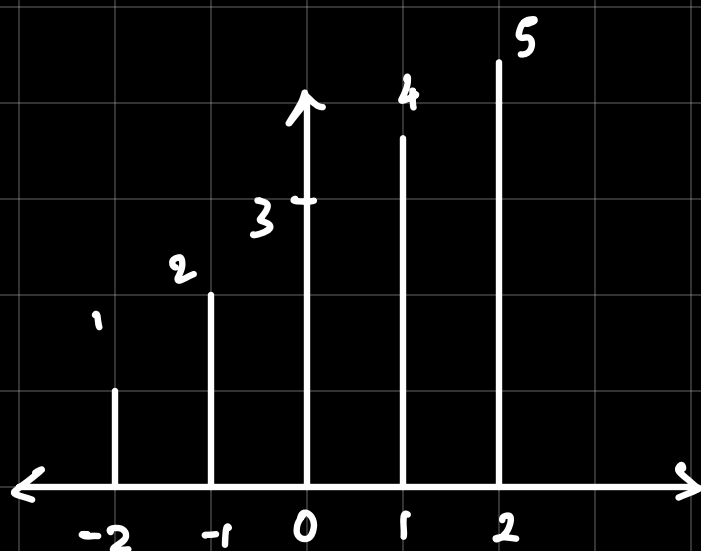
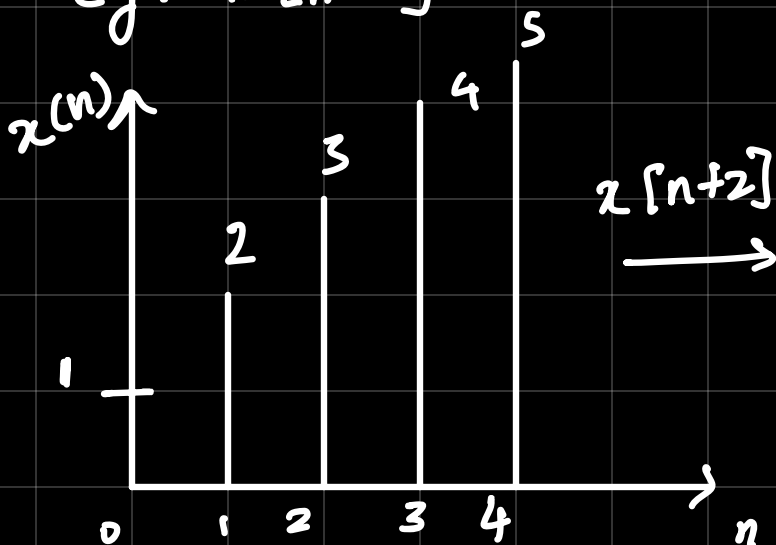
$\uparrow \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6$

## ② Time Advance Operation / Time Shifting

$$x[n+k]$$

↳ advance by  $k$  units → Shift left

eg:  $x[n+2]$

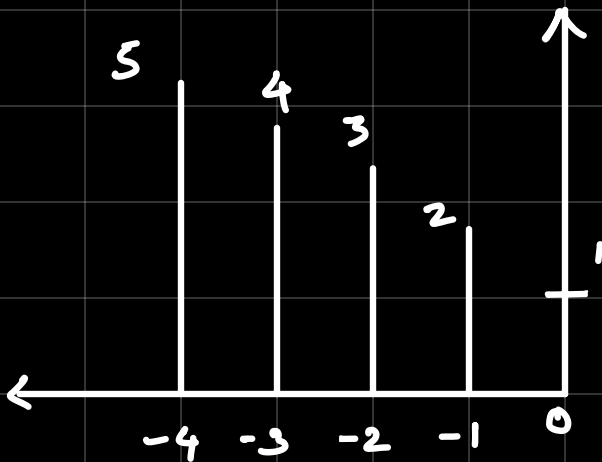


$$x[n+2] = \left\{ \begin{array}{ccccc} 1, & 2, & 3, & 4, & 5 \\ -2 & -1 & \uparrow & 1 & 2 \end{array} \right\}$$

### ③ Time folding / Mirror / Time Reversed

$$x[-n]$$

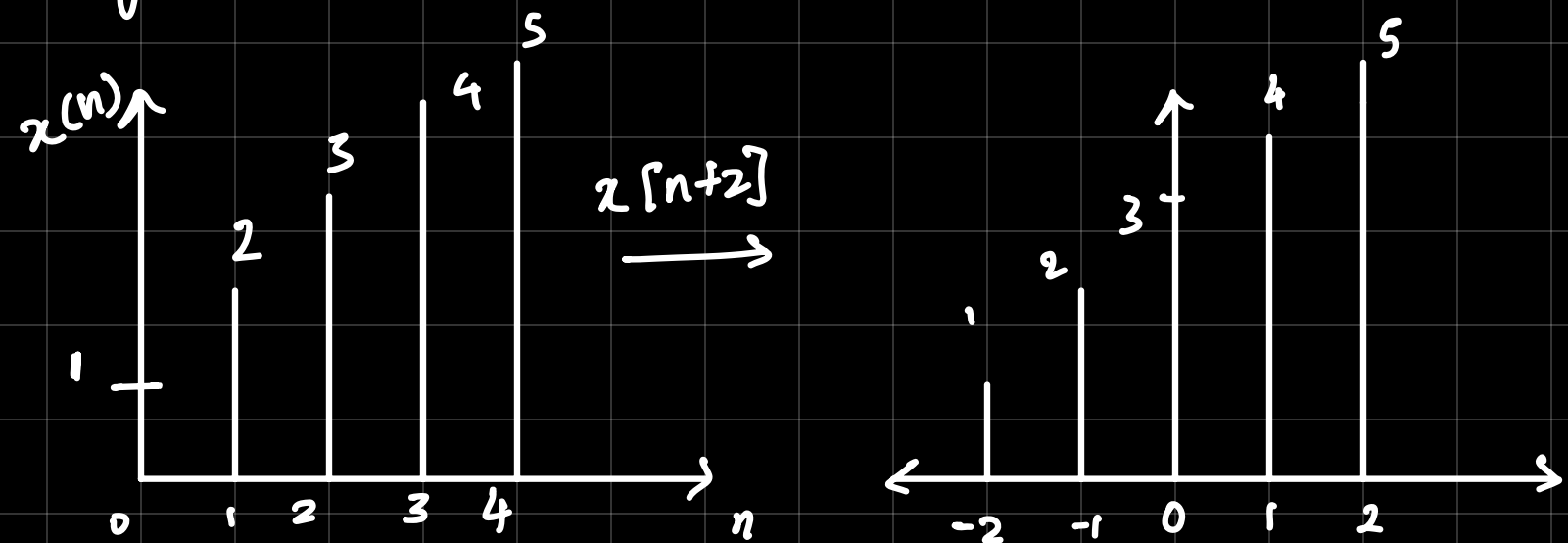
eg: (contd.)

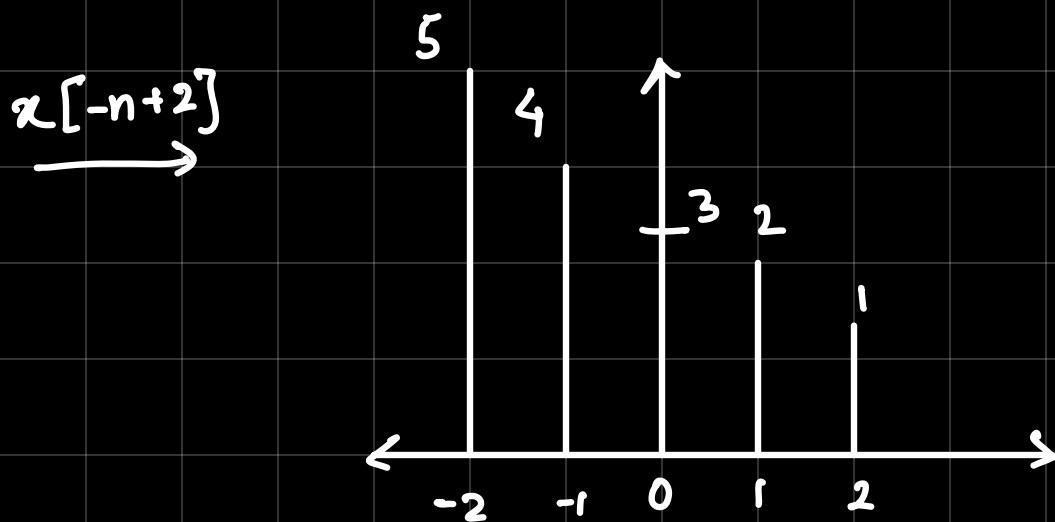


### ④ Folding & Delay

$$x[-n+k] \quad \swarrow \text{advance} \Rightarrow \text{fold}$$

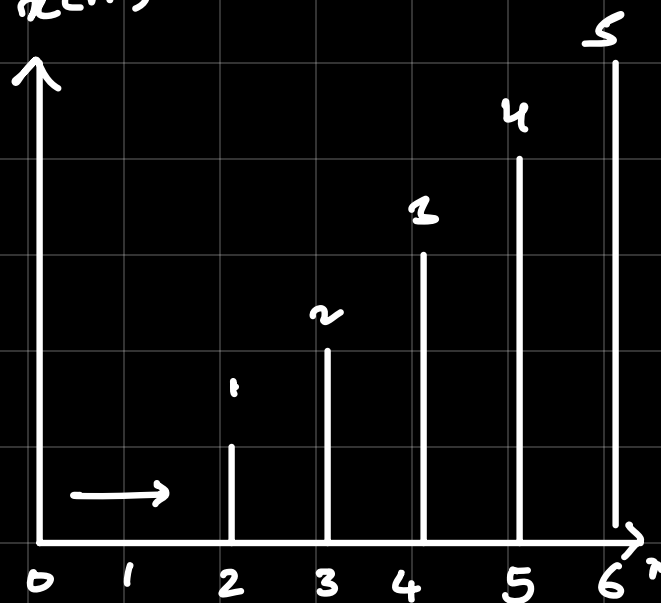
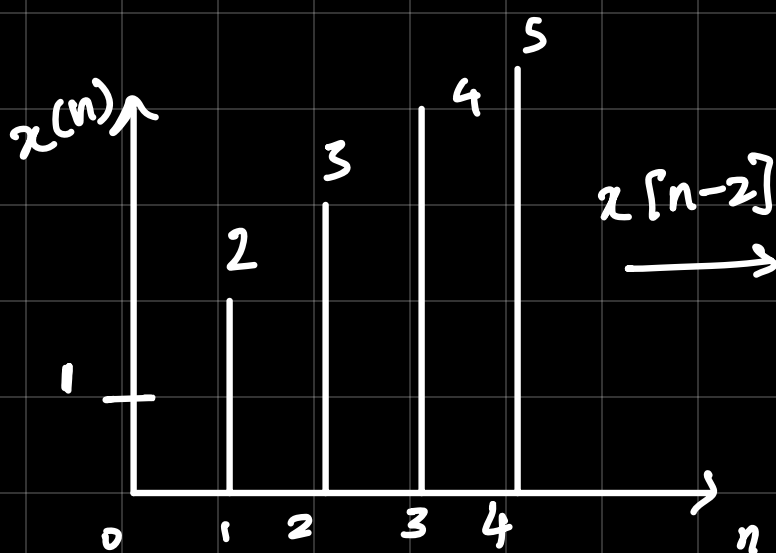
eg:  $x[-n+2]$

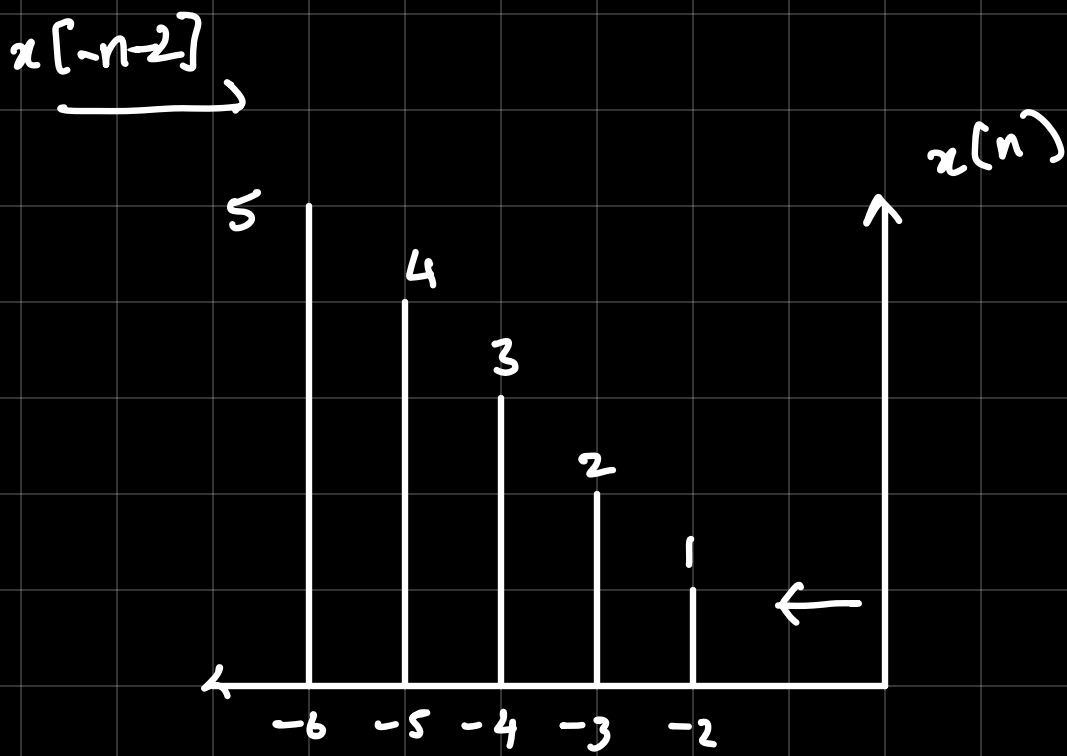




## ⑤ Folding & Advance

$x[-n-k] \Rightarrow$  delay  $\rightarrow$  fold





⑥ Time Scaling — Compression (down scaling)  
 — Expansion (up scaling)

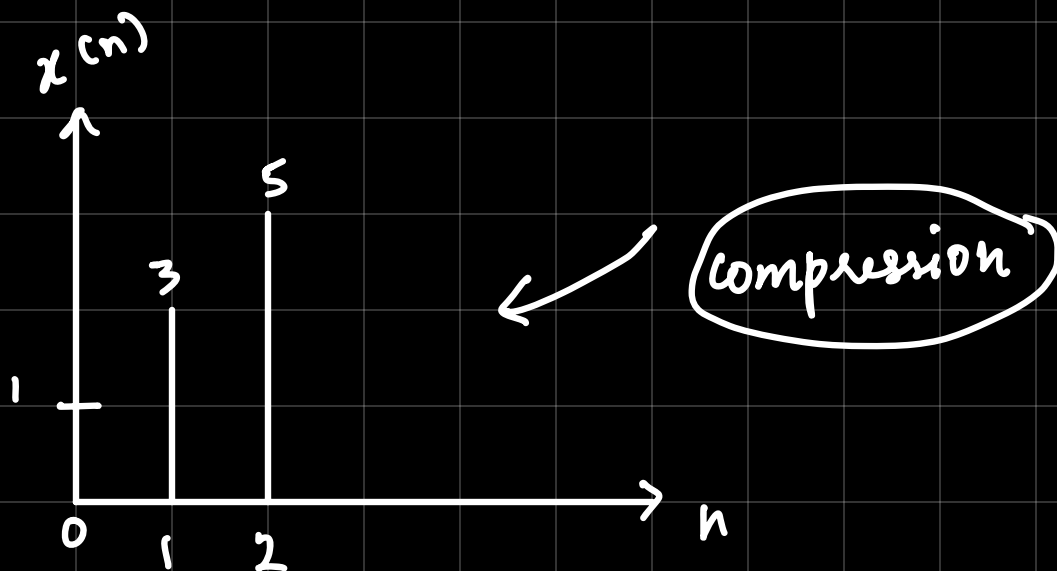
$$x[n] = \begin{bmatrix} 1, 2, 3, 4, 5 \\ 0 \quad 1 \quad 2 \quad 3 \quad 4 \end{bmatrix}$$

$$\hookrightarrow x[kn] \rightarrow k=2 \rightarrow x[2n]$$



$n$        $x[kn]$       amp. at  $x[n]$

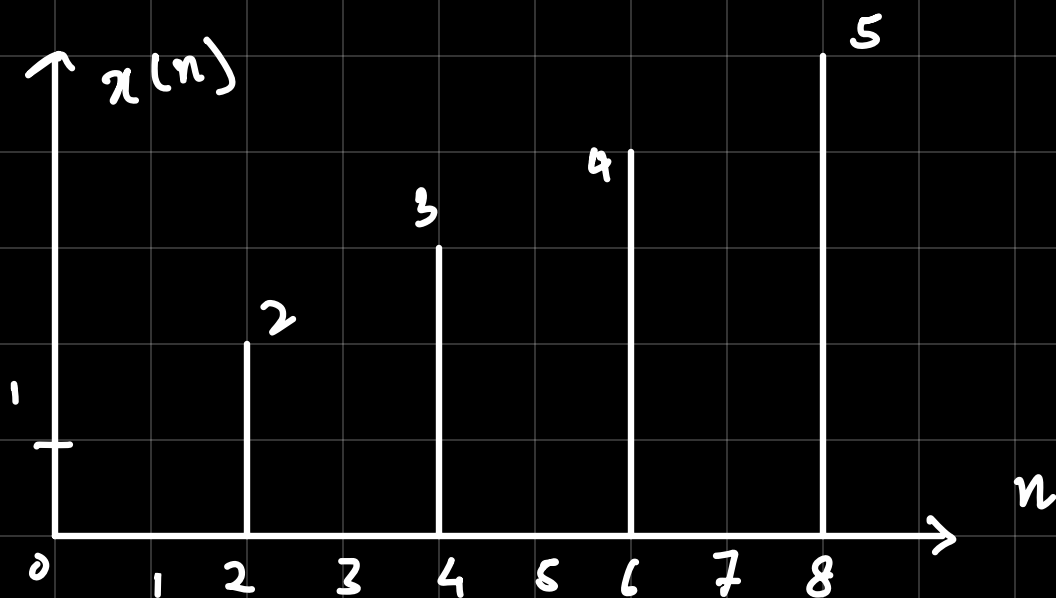
|        |     |         |     |          |
|--------|-----|---------|-----|----------|
| $y(0)$ | $=$ | $x[0]$  | $=$ | 1        |
| $y(1)$ | $=$ | $x[2]$  | $=$ | 3        |
| $y(2)$ | $=$ | $x[4]$  | $=$ | 5        |
| $y(3)$ | $=$ | $x[6]$  | $=$ | $\infty$ |
| $y(4)$ | $=$ | $x[8]$  | $=$ | $\infty$ |
| $y(5)$ | $=$ | $x[10]$ | $=$ | $\infty$ |



$$\hookrightarrow x[n/k] \rightarrow k=2 \rightarrow x[n/2]$$

|        |     |          |     |          |
|--------|-----|----------|-----|----------|
| $y(0)$ | $=$ | $x[0]$   | $=$ | 1        |
| $y(1)$ | $=$ | $x[1/2]$ | $=$ | $\infty$ |
| $y(2)$ | $=$ | $x[1]$   | $=$ | 2        |
| $y(3)$ | $=$ | $x[3/2]$ | $=$ | $\infty$ |

$$\begin{aligned}
 y(4) &= x[2] = 3 \\
 y(5) &= x[5/2] = x \\
 y(6) &= x[3] = 4 \\
 y(7) &= x[7/2] = x \\
 y(8) &= x[4] = 5
 \end{aligned}$$



② Amplitude Scaling — Amplification / Attenuation

Amplification

$$y[n] = 2x[n]$$

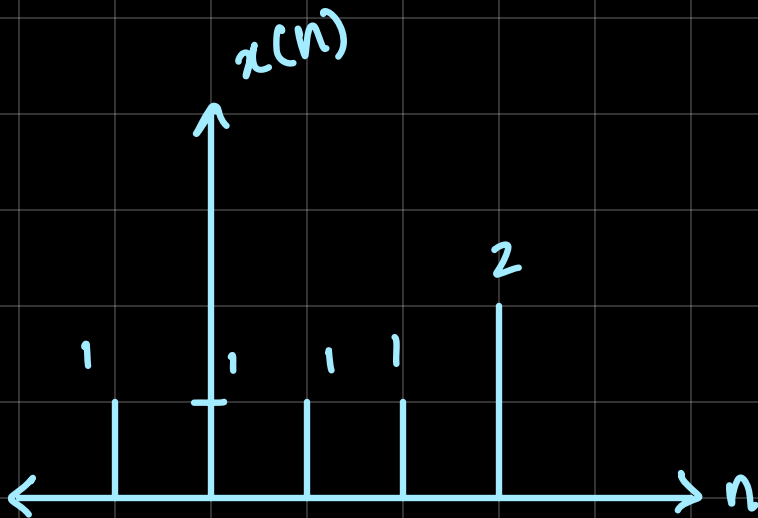
←  $\neq x[2n]$  ← time scaling

Attenuation

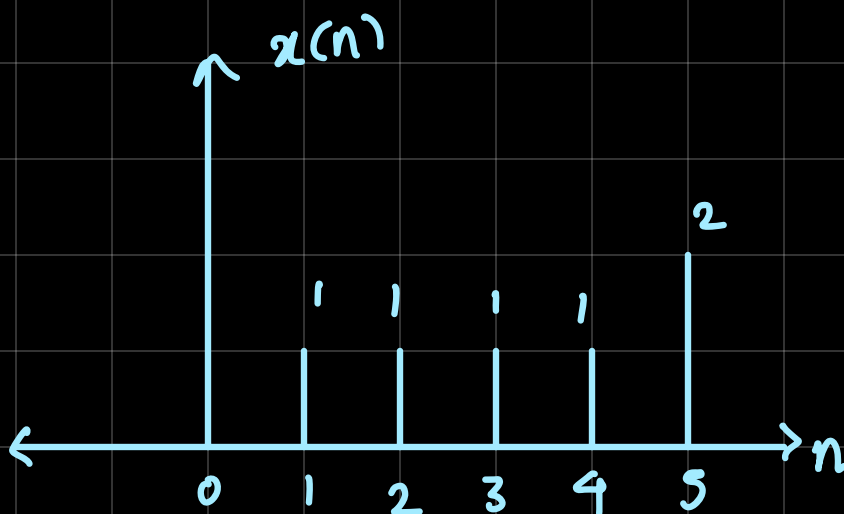
$$y[n] = \frac{1}{2} x[n]$$

## Questions

Q1. a)  $x[n] = \{1, 1, 1, 1, 2\}$

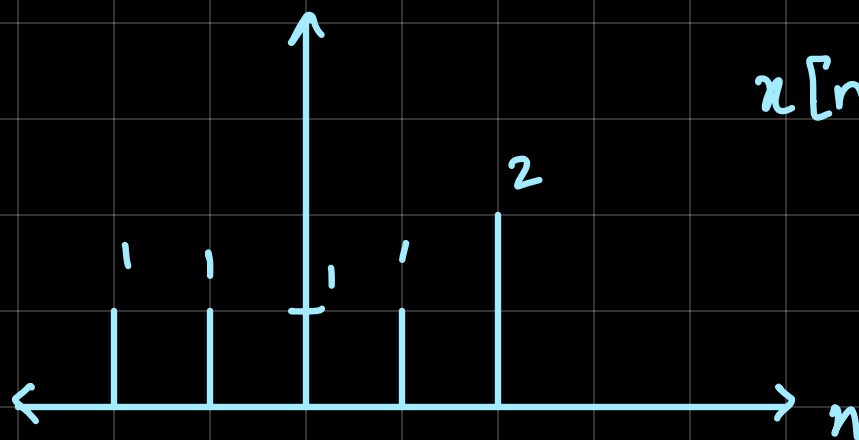


b)  $x[n-2]$   
↳ delay



$x[n-2] = \{0, 1, 1, 1, 1, 2\}$

c)  $x[n+1]$   
 $\hookrightarrow$  advance  
 $x(n)$

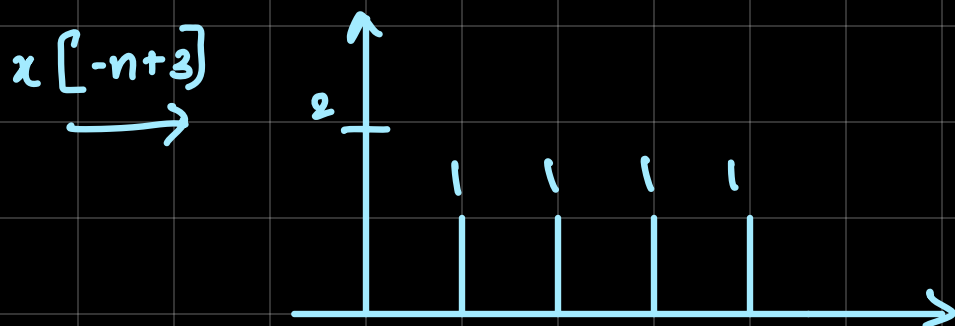
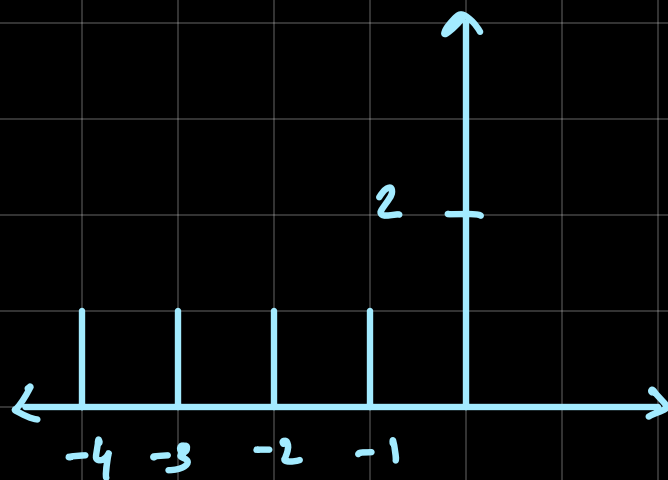
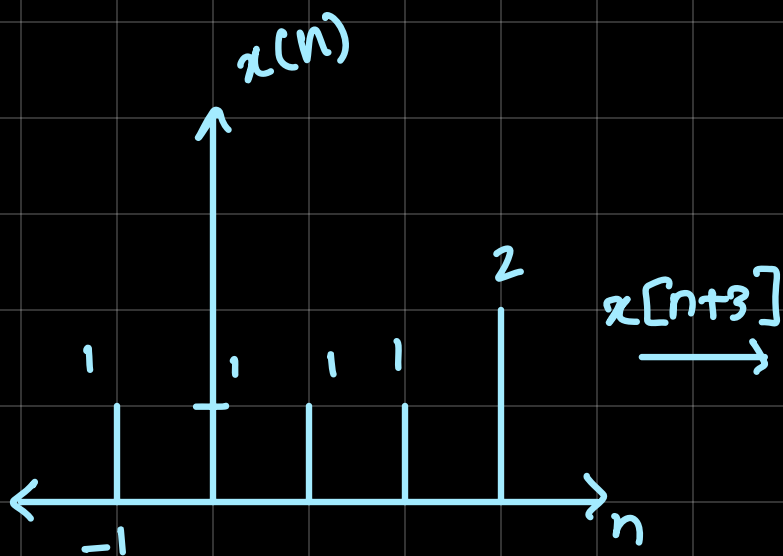


$$x[n+1] = \{1, 1, 1, 1, 2\}$$

↑

d)  $x(3-n) = x[-n+3]$

$\hookrightarrow$  advance & fold

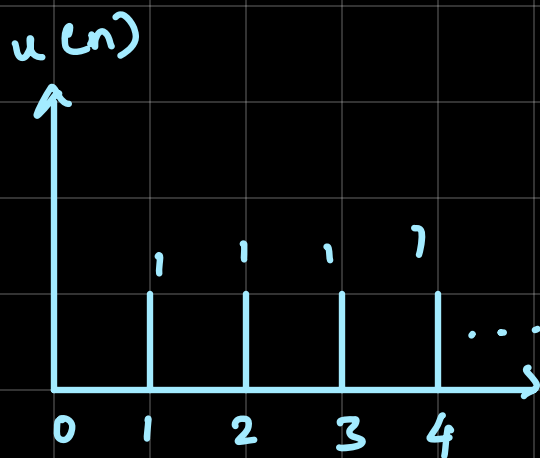
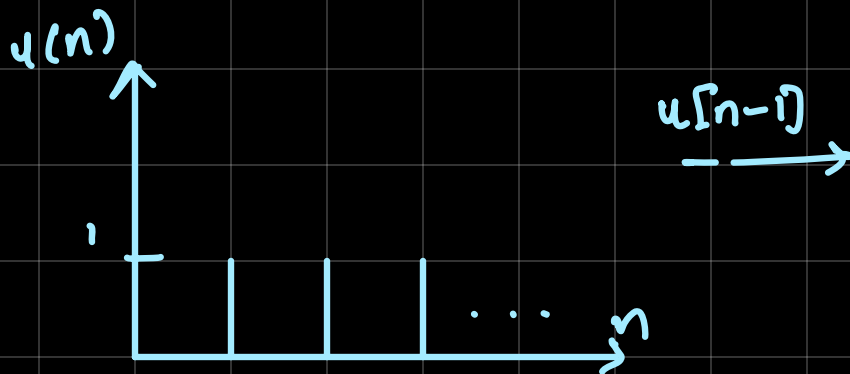


$$x[-n+3] = \{2, 1, 1, 1, 1\}$$

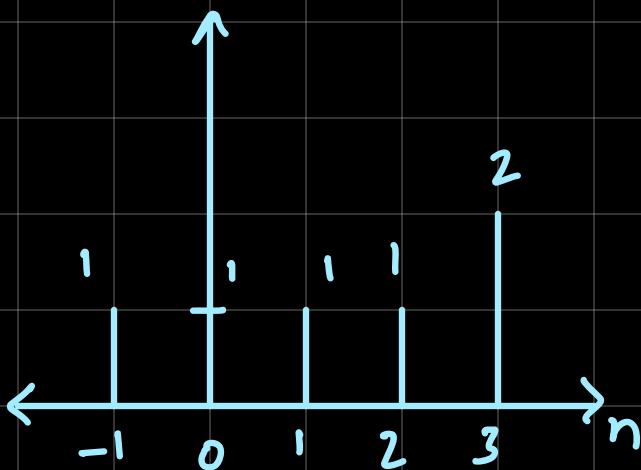
↑

$$e) x[n] \cdot u[n-1]$$

$$u[n] = 1 \quad n \geq 0$$

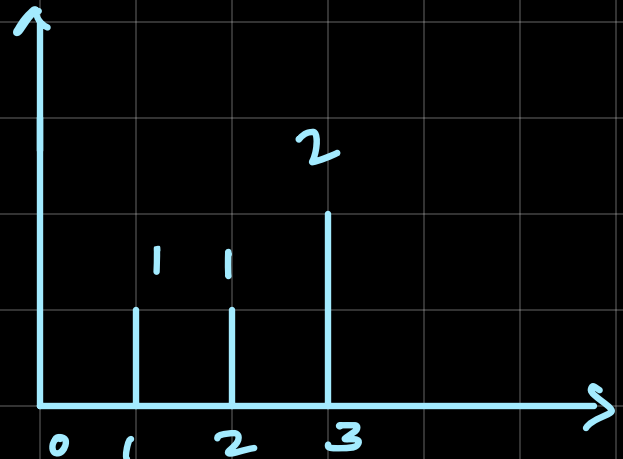


$x[n]$

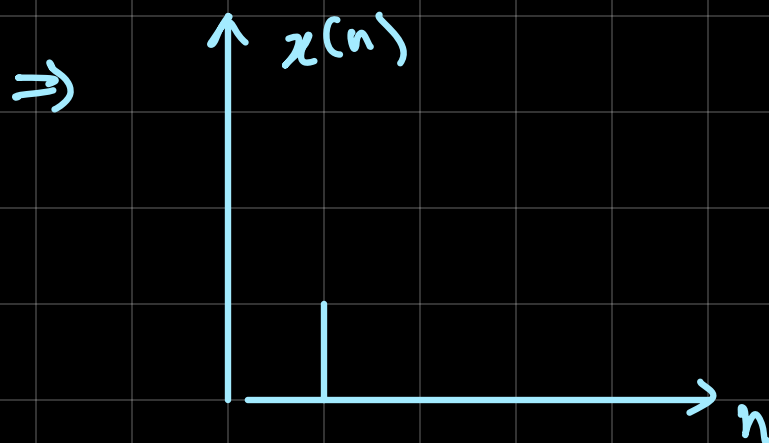
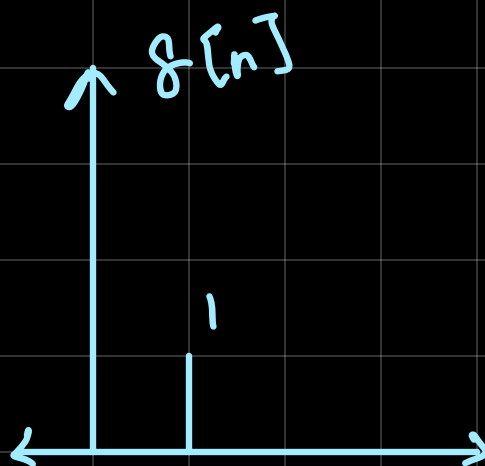
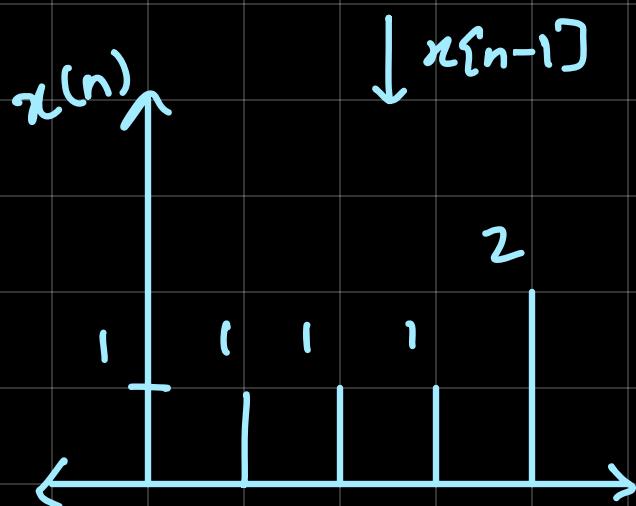
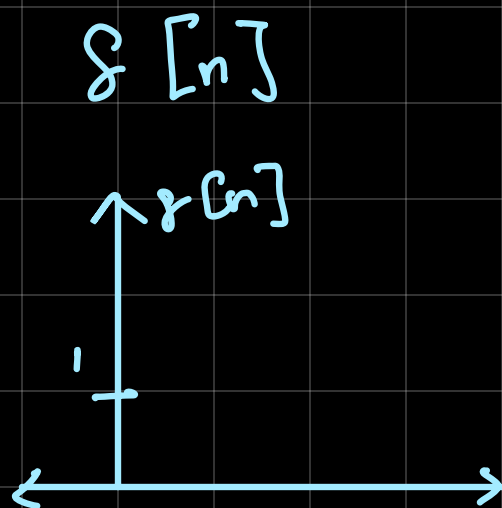
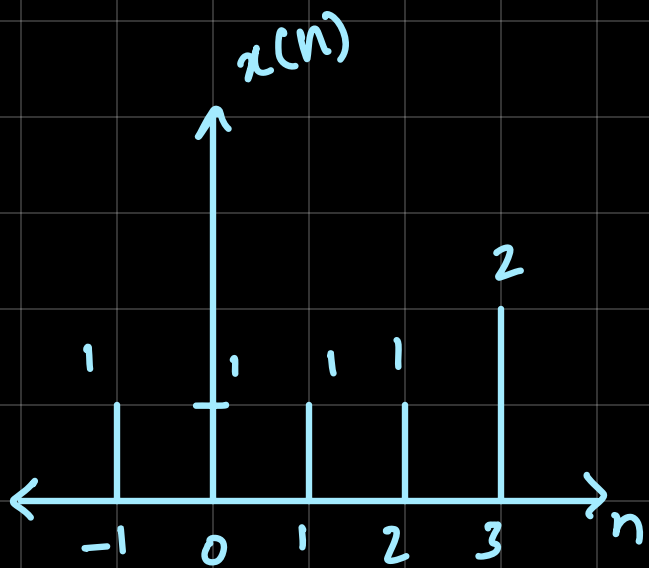


$$u[n-1] \cdot x[n] = \{0, 1, 1, 2\}$$

↑



f)  $x[n-1] \otimes g[n-1]$



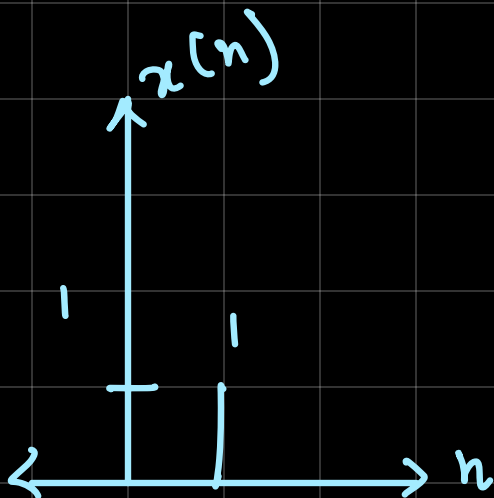
$y[n] = \{0, 1\}$

g) Even samples of  $x[n]$

↳  $x[2n]$

⇒ compression

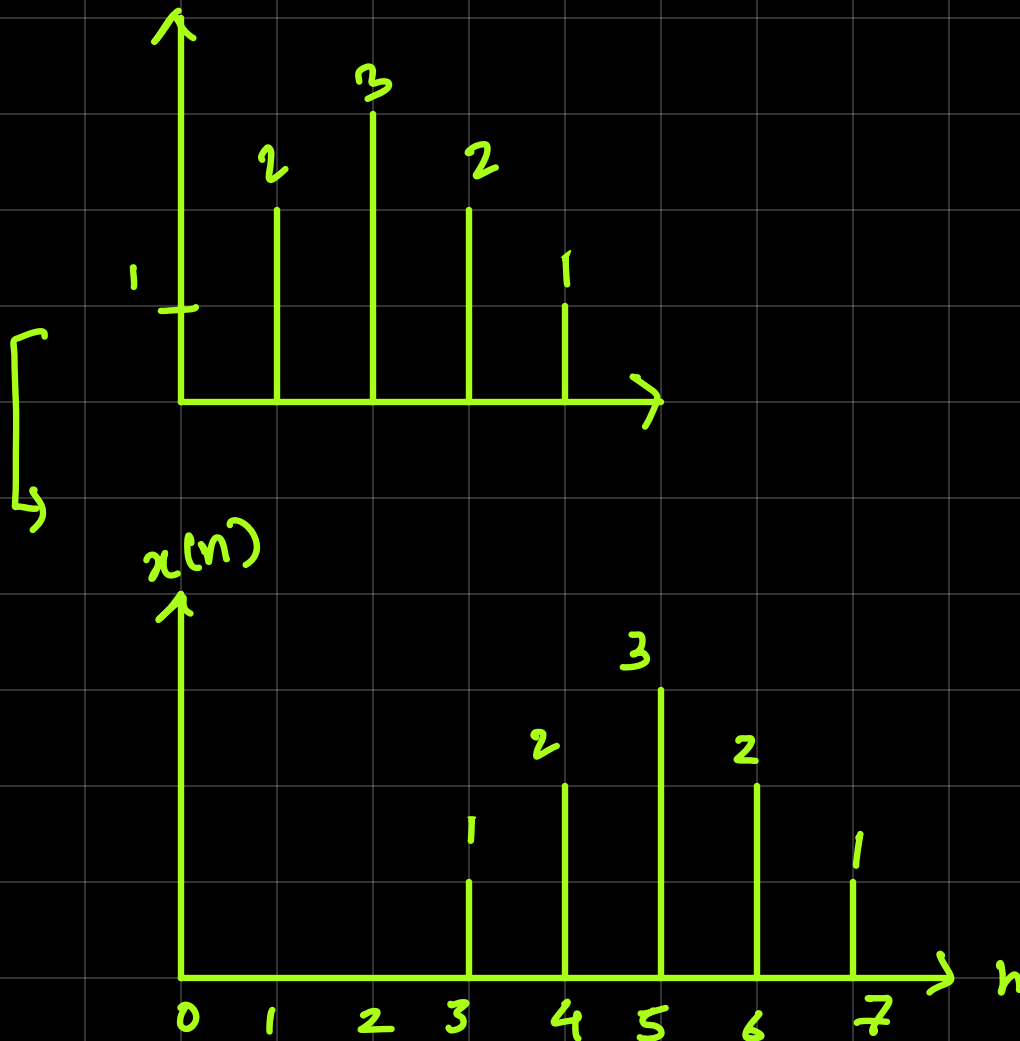
$$\begin{aligned} y(0) &= x[0] = 1 \\ y(1) &= x[2] = 1 \\ y(2) &= x[4] = 1 \\ y(3) &= x[6] = 1 \\ y(4) &= x[8] = 1 \\ y(5) &= x[10] = 1 \end{aligned} \quad \left. \vphantom{\begin{aligned} y(0) &= x[0] = 1 \\ y(1) &= x[2] = 1 \\ y(2) &= x[4] = 1 \\ y(3) &= x[6] = 1 \\ y(4) &= x[8] = 1 \\ y(5) &= x[10] = 1 \end{aligned}} \right\} x$$



Q.  $x[n] = \{1, 2, 3, 2, 1\}$

↑

a)  $x[n-3]$

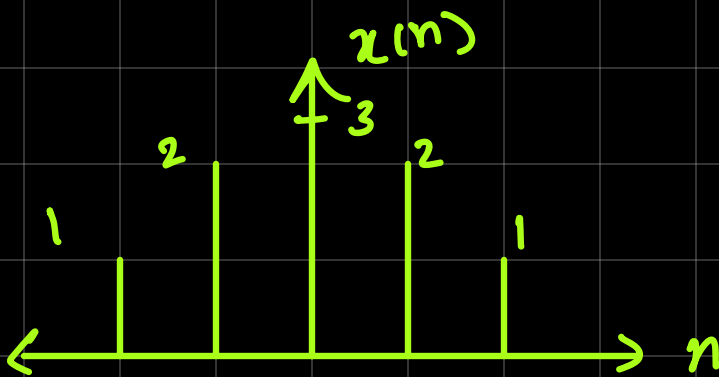


$x[n-3] = \{0, 0, 0, 1, 2, 3, 2, 1\}$

↑



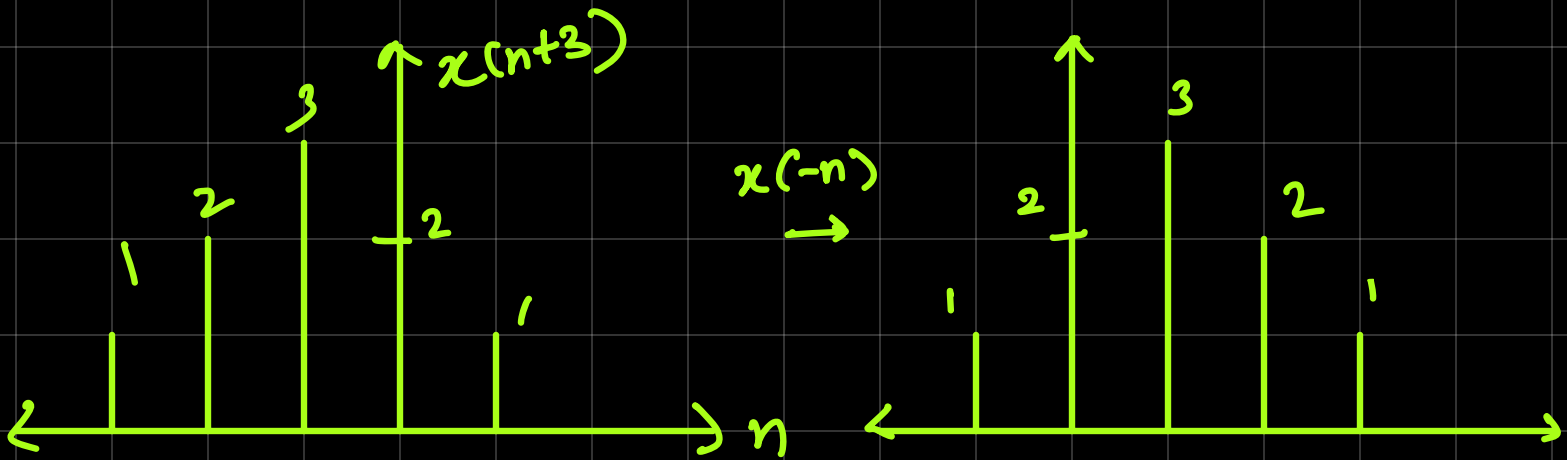
b)  $x(n+2)$



$$x(n+2) = \{1, 2, 3, 2, 1\}$$

↑

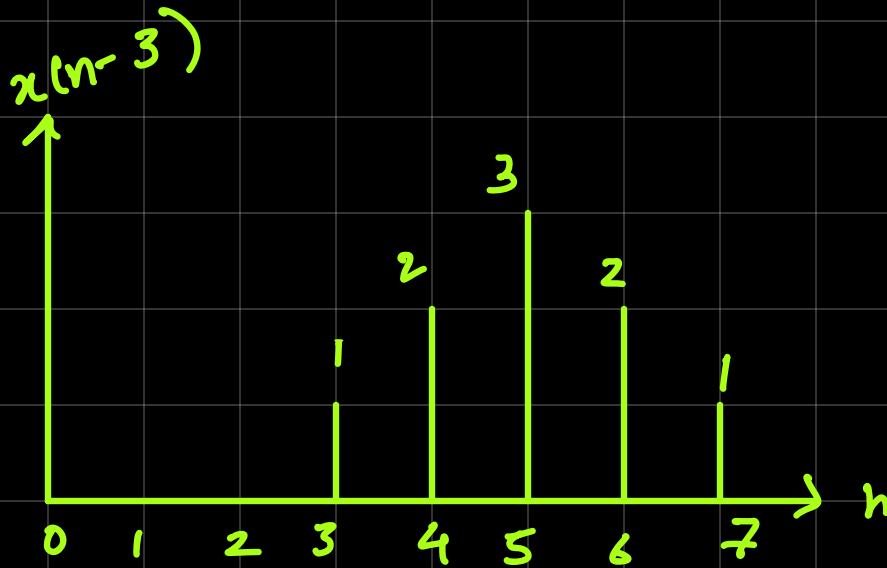
c)  $x(-n+3)$



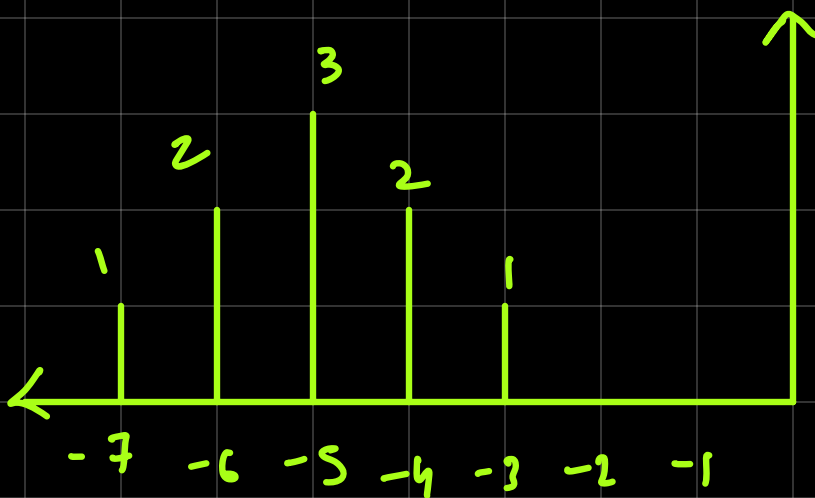
$$x(-n+3) = \{1, 2, 3, 2, 1\}$$

↑

d)  $x(-n-3)$



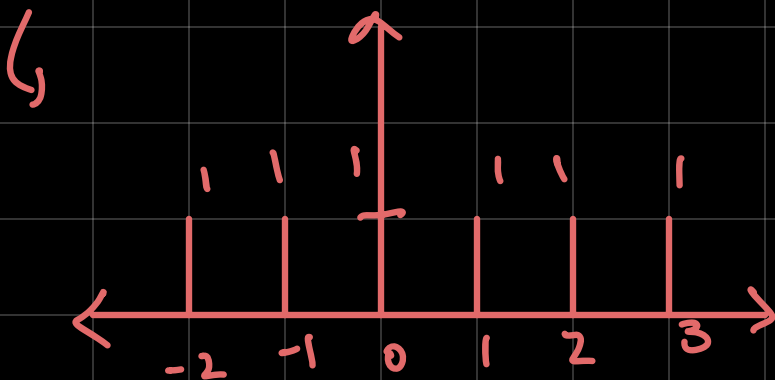
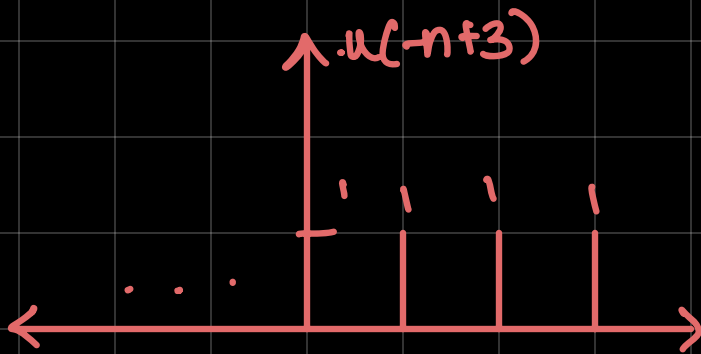
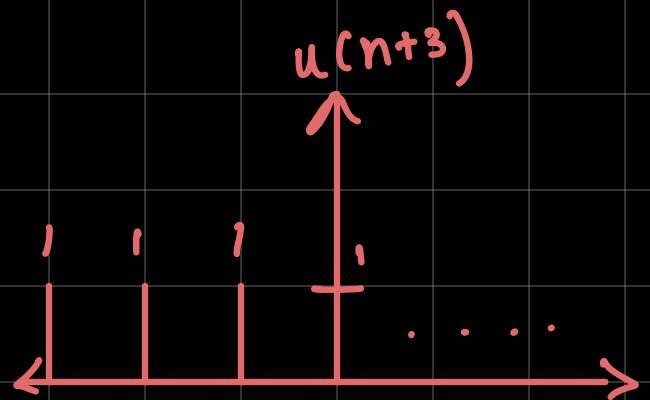
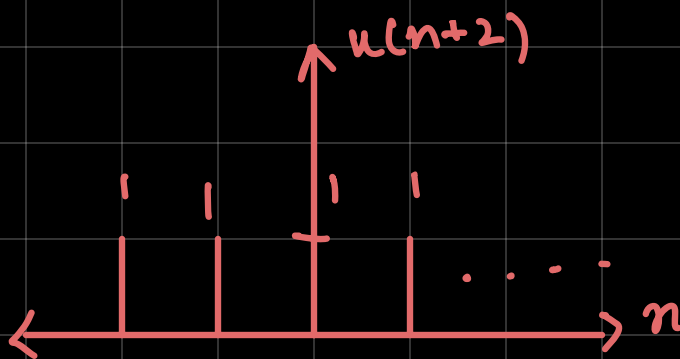
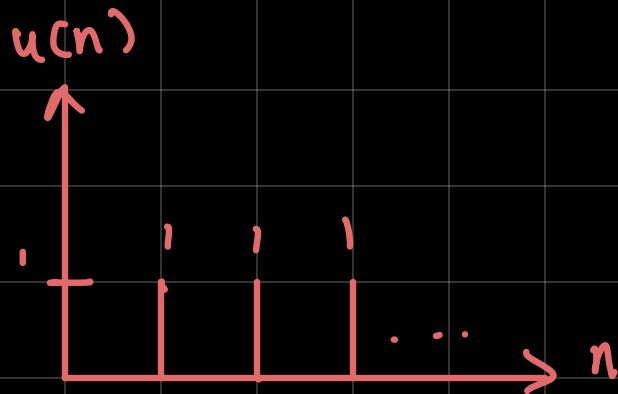
$\downarrow x(-n)$



$$x[-n-3] = \{1, 2, 3, 2, 1, 0, 0, 0\}$$

↑

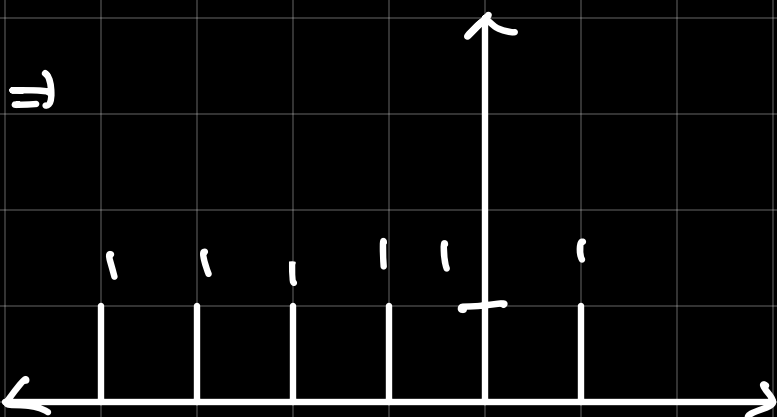
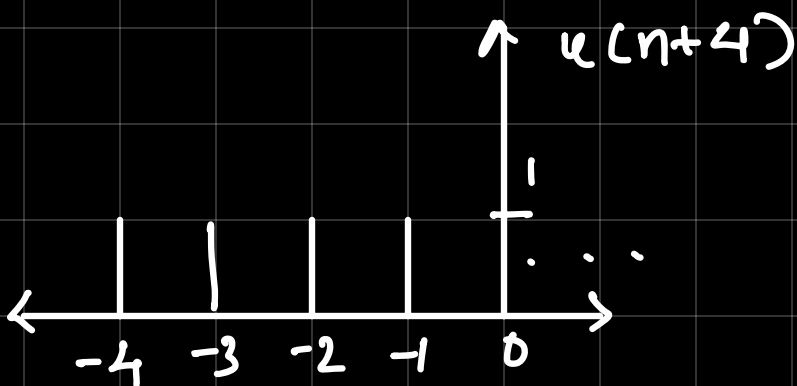
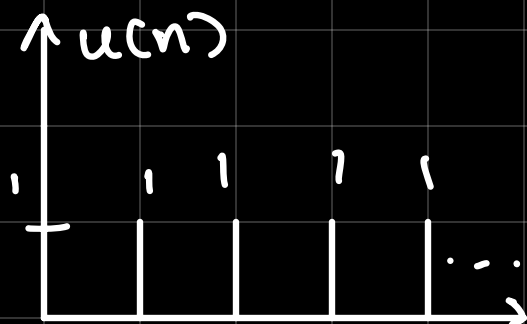
Q.  $u(n+2)u(-n+3)$



$$y[n] = \{1, 1, 1, 1, 1, 1\}$$

$\uparrow$

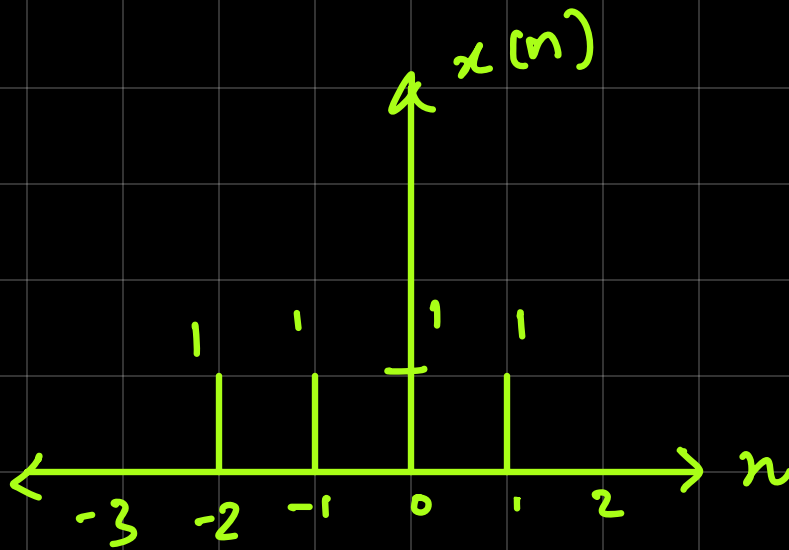
Q.  $x[n] = u[n+4] - u[n-2]$



$$x[n] = \{1, 1, 1, 1, 1, 1\}$$

↑

Q.  $x(n) = \delta(n+2) + \delta(n+1) + \delta(n) + \delta(n-1)$



$$x(n) = \begin{cases} 0 & n \leq -3 \\ 1 & -2 \leq n \leq 1 \\ 0 & n \geq 2 \end{cases}$$

# Classification of DTS

- ① Deterministic & Random
- ② Periodic & Non-Periodic
- ③ Energy & Power
- ④ Causal & Non-causal
- ⑤ Even & odd

## Deterministic

- Exhibiting no uncertainty of its magn. & phase at any instant → deterministic
  - Can be represented completely by mathematical eqn. at any time & its nature, & amp. can be predicted.
- Eg: sinusoidal, exponential, ramp.

## Periodic & Non-Periodic

- A signal which has a def. pattern & repeats itself at regular intervals of time.
- A signal which does not repeat at regular intervals is called aperiodic

$$\rightarrow x(n) \rightarrow \text{periodic} \Rightarrow x(n) = x(n+N)$$

↙  
smallest  
value =  
fund.  
period

↙ angular frequency

$$\rightarrow \omega = 2\pi/N$$

$$N = 2\pi/\omega$$

Q. Check if periodic or not

$$x(n) = A \cos [\omega_0 n + \theta]$$

$$x[n+N] = x[n]$$

$$\Rightarrow A \cos [\omega_0 (n+N) + \theta]$$

$$\Rightarrow A \cos [2\pi f_0 n + 2\pi f_0 N + \theta]$$

$$f_0 = \frac{k}{N} \left. \begin{array}{l} \leftarrow \text{integer} \\ \leftarrow \text{integer} \end{array} \right\} \text{rational}$$

$\therefore$  periodic



$$Q. x(n) = e^{j\omega_0 n}$$

$$\hookrightarrow x(n+N) = x(n)$$

$$e^{j\omega_0(n+N)} = e^{j\omega_0 n}$$

$$\frac{e^{j\omega_0 n + j\omega_0 N}}{e^{j\omega_0 n}} = \frac{e^{j\omega_0 N}}{1} = e^{j\omega_0 N}$$

$$e^{j\omega_0 N} = 1$$

$$e^{j\omega_0 N}$$

=

$$e^{j\theta}$$

$2\pi k$

$$e^{j2\pi k} = 1$$

$$j\omega_0 N = j2\pi k \rightarrow \frac{\omega_0}{2\pi} = \frac{k}{N}$$

$$2\pi f_0 N = 2\pi k$$

$$f_0 = \frac{k}{N} \left\{ \begin{array}{l} \leftarrow \text{integer} \\ \leftarrow \text{integer} \end{array} \right.$$

} rational

Q.  $\sin(0.02\pi n)$  & find  $N$ ?

$$\sin(0.02\pi n)$$

$$\downarrow \sin(\omega n) \leftarrow$$

$$\omega = 0.02\pi$$

$$2\pi f = 0.02\pi$$

$$f = 0.01$$

$$N = 100$$

Q.  $\sin(5\pi n)$

$$\sin(\omega n) = \sin(5\pi n)$$

$$\omega n = 5\pi n$$

$$2\pi f = 5\pi$$

$$f = \frac{K}{N} = \frac{5}{\textcircled{2}}_{N=2}$$

Q.  $\cos(4n)$

$$2\pi f n = 4n$$

$$\pi f = 2$$

$$f = \frac{2}{\pi}$$

← irrational

Q.  $\sin \frac{2\pi n}{3} + \cos \frac{2\pi n}{5}$

$$\frac{2\pi}{3} = 2\pi f_1$$

$$f_1 = \frac{1}{3}$$

$$N_1 = 3$$

$$\frac{2\pi}{5} = 2\pi f_2$$

$$f_2 = \frac{1}{5}$$

$$N_2 = 5$$

(periodic)

fundamental period of both signal

$$\text{LCM}(3, 5) = \underline{\underline{15}} < N$$

$$Q. \cos\left(\frac{n}{6}\right) \cos\left(\frac{n\pi}{6}\right)$$

aperiodic  $\times$  periodic = aperiodic

$$Q. \cos\left(\frac{\pi}{2} + 0.3n\right)$$

$$\hookrightarrow \sin(0.3n)$$

$$0.3x = 2\pi + x$$

$$f = \frac{3}{2\pi}$$

non-periodic

# Symmetric & Asymmetric

Even signal :  $x(n) = x(-n)$

↑  
fold

Odd signal :  $x(-n) = -x(n)$

$$x(n) = x_o(n) + x_e(n)$$

↑  
odd part

↑  
even part

$$x_e(n) = \frac{x(n) + x(-n)}{2}$$

$$x_o(n) = \frac{x(n) - x(-n)}{2}$$

$$a) \quad x(n) = 3^n$$

$$x(-n) = 3^{-n}$$

$$x_e(n) = \frac{3^n + 3^{-n}}{2} = \frac{1}{2} [3^n + 3^{-n}]$$

$$x_o(n) = \frac{3^n - 3^{-n}}{2} = \frac{1}{2} [3^n - 3^{-n}]$$

$$b) \quad x(n) = 3 e^{j \frac{5\pi n}{2}} \rightarrow 3 e^{j\theta}$$

$$\begin{aligned} 3 e^{j\theta} &= 3 [\cos \theta + j \sin \theta] \\ &= 3 \left[ \cos \frac{5\pi n}{2} + j \sin \frac{5\pi n}{2} \right] \end{aligned}$$

$$x(-n) = 3 \left[ \cos \frac{5\pi n}{2} - j \sin \frac{5\pi n}{2} \right]$$

even part:  $\frac{1}{2} [x(n) + x(-n)]$

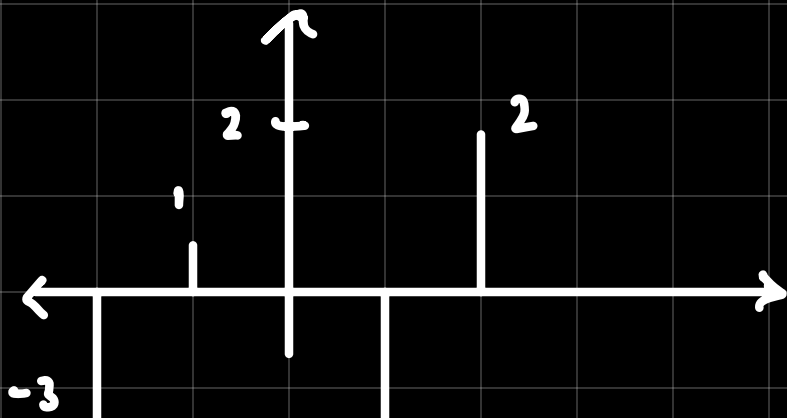
$$= \frac{1}{2} \times 6 \cos \frac{5\pi n}{2} = 3 \cos \frac{5\pi n}{2}$$

odd part:

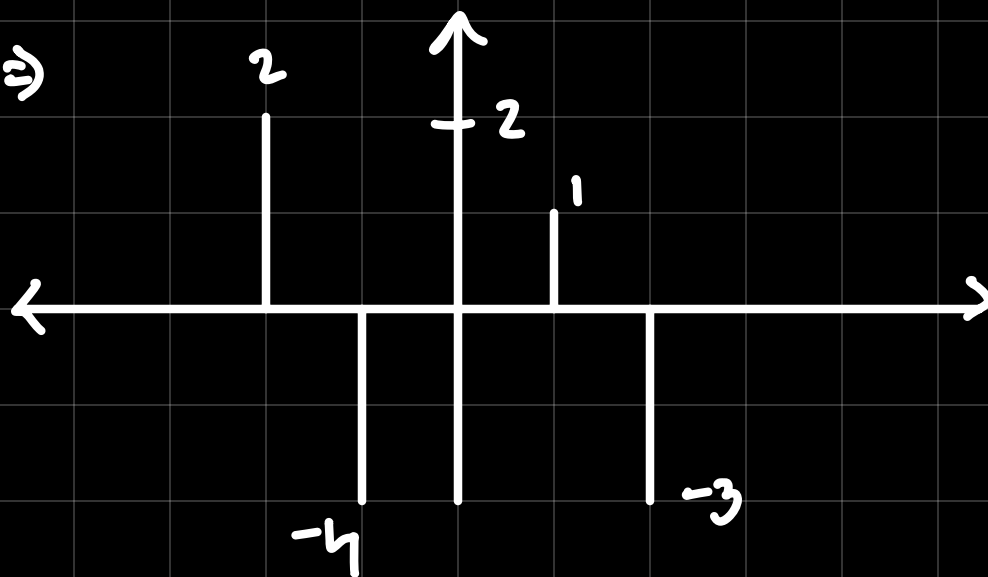
c)  $x(n) = \{-3, 1, 2, -4, 2\}$

even:  $\frac{1}{2} [x(n) + x(-n)]$

$x(n) \Rightarrow$



$x(-n) \Rightarrow$



$$\Rightarrow \frac{1}{2} [-1, -3, 4, -1]$$

$$x_e[n] = [-0.5, -1.5, \underset{\uparrow}{2}, -0.5]$$

$$x_o[n] \Rightarrow \frac{1}{2} [-5, 5, 0, -5, 5]$$

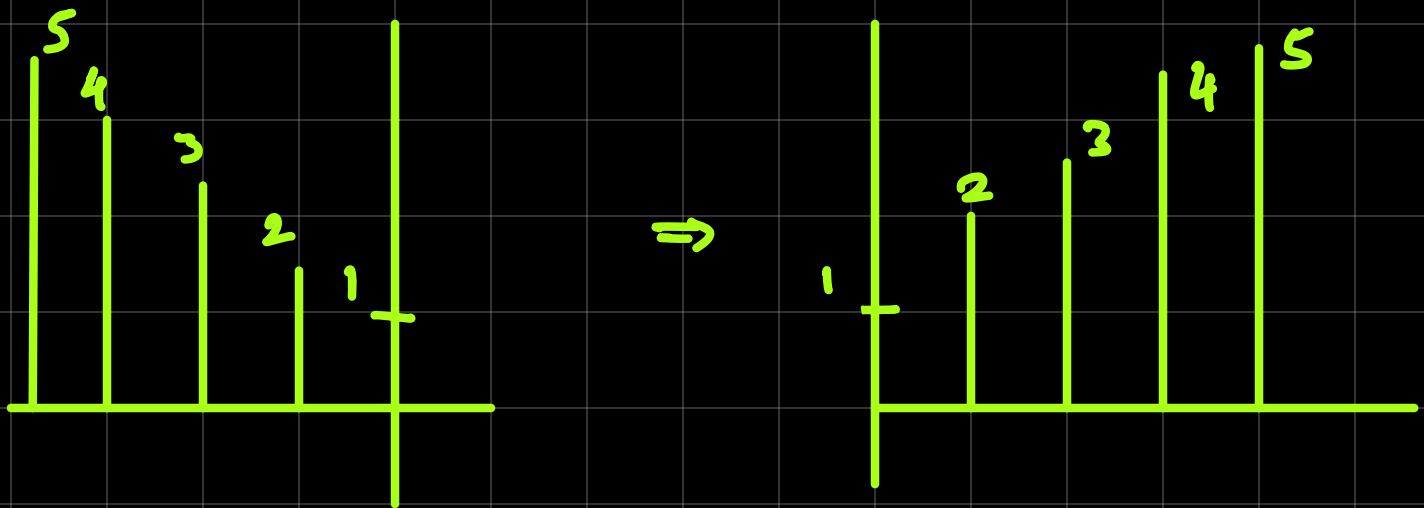
$$\Rightarrow [-2.5, 2.5, \underset{\uparrow}{0}, -2.5, 2.5]$$



$$d) \quad x(n) = \{5, 4, 3, 2, 1\}$$

↑

even:  $\frac{1}{2} [x(n) + x(-n)]$



$$x_e[n] = \frac{1}{2} [5, 4, 3, 2, 2, 2, 3, 4, 5]$$

$$= [2 \cdot 5, 2, 1 \cdot 5, 1, 1, 1, 1 \cdot 5, 2, 2 \cdot 5]$$

↑

$$x_o[n] = \frac{1}{2} [5, 4, 3, 2, 0, 2, 3, 4, 5]$$

$$= [2 \cdot 5, 2, 1 \cdot 5, 1, 0, 1, 1 \cdot 5, 2, 2 \cdot 5]$$

↑

# Causal, Non-Causal & Anti-causal signals

Causal :

$$x(n) = 0 \quad \text{for } n < 0$$

eg: unit step, unit ramp

Non-causal: can exist for both  $n > 0$  &  $n < 0$

$$x(n) \neq 0 \quad \text{for } n < 0$$

Eg: Exponential signal

Anti-causal:

$$x(n) = 0 \quad \text{for } n \geq 0$$

# Energy & Power Signals

Discrete:

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 \quad 0 < E < \infty$$

$$P = 0$$

Periodic

$$P = \frac{E}{N} = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

Q.  $x(n) = 3(0.5)^n u(n)$   
↳ aperiodic

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

↳  $u(n) \Rightarrow n \geq 0$

$$= \sum_{n=0}^{\infty} |x(n)|^2$$

$$= \sum_{n=0}^{\infty} (3 (0.5)^n)^2$$

$$= 9 \left( \frac{1}{1-(0.5)^2} \right)^2$$

$$= \frac{9}{0.75} = 12$$

$P=0 \Rightarrow$  because energy is finite & aperiodic

Q.  $x(n) = 6 \cos\left(\frac{2\pi n}{4}\right) \quad N=4$

$$\text{Power} = \frac{1}{N} \sum_{n=0}^3 |x(n)|^2$$

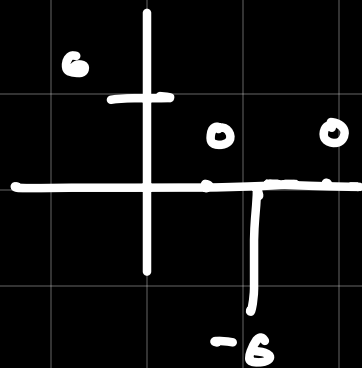
$$= \frac{1}{4} \sum_{n=0}^3 \left[ 6 \cos\left(\frac{2\pi n}{4}\right) \right]^2$$

$$= \frac{1}{4} (36 + 0 + 36 + 0)$$

$$= \underline{\underline{18W}}$$

$$x(n) = [6, 0, -6, 0]$$

↑



$$Q. \quad x(n) = \left(\frac{1}{2}\right)^n \mu(n)$$

↳ aperiodic

$$E = \sum_{n=0}^{\infty} |x(n)|^2$$

↳  $\mu(n)$

$$= \sum_{n=0}^{\infty} \left[\left(\frac{1}{2}\right)^n\right]^2$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{2n}$$

$$= 1 + \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \frac{1}{2^8}$$

$$= \frac{1}{1 - \left(\frac{1}{2}\right)^2} = \frac{4}{3} \text{ J}$$

$$* \sum_{n=0}^N \alpha^n = \frac{1 - \alpha^{N+1}}{1 - \alpha}$$

$$Q \quad x(n) = e^{j(\frac{\pi}{3}n + \frac{\pi}{2})}$$

$$\downarrow$$

$$\cos\left(\frac{\pi}{3}n + \frac{\pi}{2}\right) + j \sin\left(\frac{\pi}{3}n + \frac{\pi}{2}\right)$$

$$\dot{E} = \sum_{-N}^N \left( \cos^2\left(\frac{\pi}{3}n + \frac{\pi}{2}\right) + j^2 \sin^2\left(\frac{\pi}{3}n + \frac{\pi}{2}\right) + 2j \right) \quad j = \sqrt{-1}$$

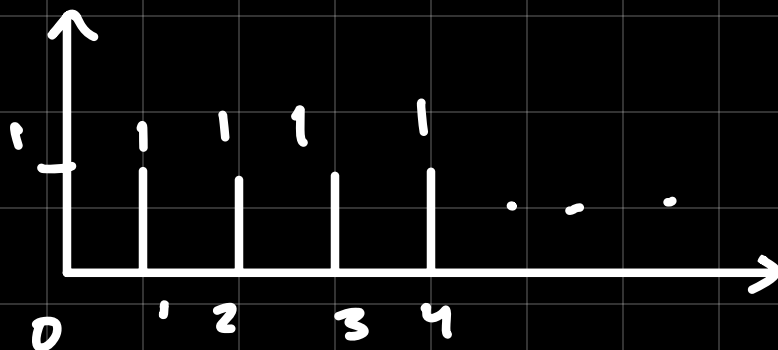
$$= \sum_{-N}^N \sum_{-N}^N \sum_{-N}^N \sqrt{\cos^2\left(\frac{\pi}{3}n + \frac{\pi}{2}\right) + \sin^2\left(\frac{\pi}{3}n + \frac{\pi}{2}\right)}$$

Q-  $x(n) = \mu(n) - \mu(n-6)$

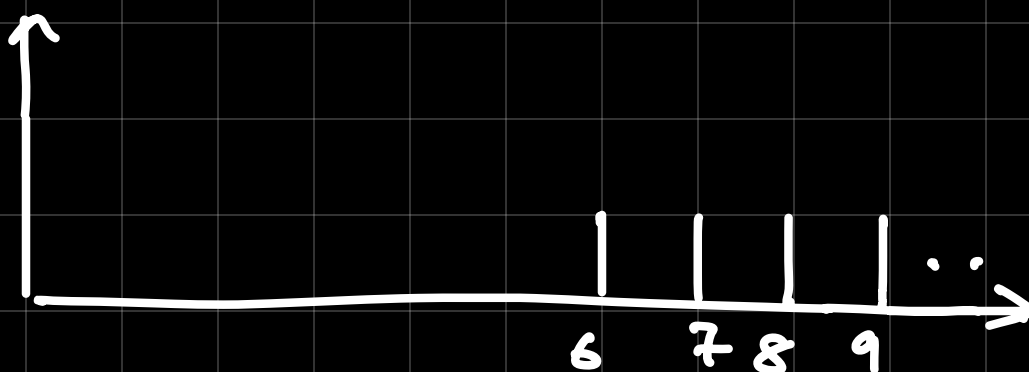
↳ aperiodic

$P=0$

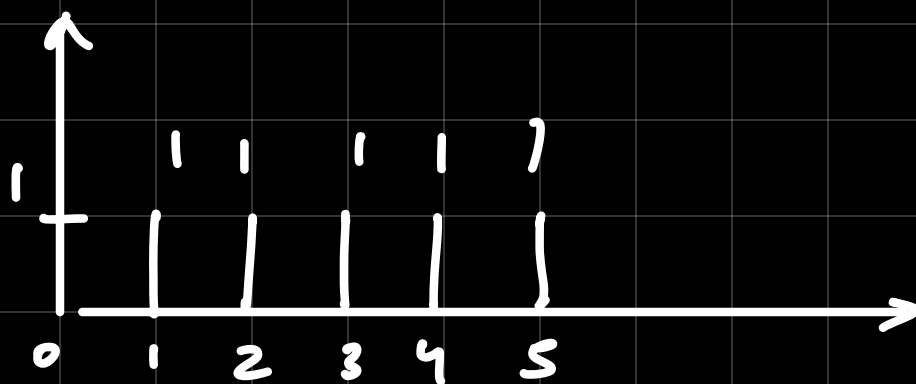
$\mu(n)$



$\mu(n-6)$



$\mu(n) - \mu(n-6)$





$$E = \sum_{n=0}^5 1 = 6$$

System



Classification of system:

- 1) Static & Dynamic
- 2) Causal & Non-causal
- 3) Time invariant & time variant
- 4) Linear & non linear
- 5) Stable & Unstable
- 6) FIR & IIR
- 7) Recursive & non-recursive

Static

↳ memoryless



o/p depends at any

instant depends on present i/p

$$y(n) = x(n)$$

Resistive ckt

Dynamic

↳ memory based



o/p depends on  
future, past i/p & o/p

(with storage)  
Inductor, Capacitor

$$y(n) = x(n) + x(n-2]$$

$$y(n) = y(n-1) + 4y(n-2)$$

↙ ↘  
past o/p

Example:

$$y(n) = a x(n)$$

$$y(n) = n x(n) + 6 x^2(n)$$

$$y(n) = x(n) + 3 x(n-1)$$

$$y(n) = \sum_{n=0}^N x(n-n)$$

$$y(n) = x^2(n) \rightarrow \text{static}$$

$$x(n^2) \rightarrow \text{dynamic}$$

Presentation:

a)  $y(n) = x(n+2)$

Dynamic System

$$n=0 \quad y(0) = x(2)$$

$$n=1 \quad y(1) = x(3)$$

$$n=-1 \quad y(-1) = y(1)$$

System is dynamic because o/p  $y(n)$  depends on future i/p.

b)  $y(n) = x^2(n)$   
Static System

$$y(2) = x^2(2)$$

$$y(0) = x^2(0)$$

$$y(-1) = x^2(-1)$$

system is static because o/p  $y(n)$  depends on present value i/p.

c)  $y(n) = x(n-2) + x(n)$   
Dynamic System

$$y(2) = x(0) + x(2)$$

$$y(0) = x(-2) + x(0)$$

$$y(-1) = x(-3) + x(-1)$$

system is dynamic because o/p  $y(n)$  depends on both past i/p & current i/p.

## Causal & Non-causal



non-anticipative



→ o/p at any instant depends only on present & past values of i/p

→  $h(n) = 0$ ,  $n < 0$

eg:  $y(n) = n x(n)$

$$y(n) = \underbrace{x(n-2)}_{\text{past}} + x(n-1) + \underbrace{x(n)}_{\text{present}}$$

past

present

$$\rightarrow y(n) = x(n) + x(n-2)$$

$$y(n) = \sin[x(n)]$$

$$y(n) = x(2n)$$

$$y(n) = x(-n)$$

Causal past + present

Causal present

Non-causal future

Non-causal

future



$$y(n) = x(n) + x(2n)$$

$$y(n) = x^2(n) +$$

$$2x(n+2)$$

$$y(n) = x(-n)$$

$$\begin{array}{lcl} y(-1) = x(1) & \} & \text{Non-causal} \\ y(0) = x(0) & \} & \text{Causal} \\ y(1) = x(-1) & \} & \end{array}$$

its non-causal for -ve values of  $n$  cause  
then it depends on future i/p

its causal for non -ve values of  $n$  cause  
past / present

## Time variant / Time invariant

↳ independent of time  
for DTS  $\Rightarrow$  shift invariance  
↓  
time shift in i/p = corresp.  
time shift o/p

Steps to identify:

① Delay i/p by  $m$  units  $\Rightarrow y_1(n) = x(n-m)$

② Delay response of system for undelayed input  
by  $m$  units  $\Rightarrow y_2(n) = x(n-m)$   
ie replace  $n$  by  $n-m$

③ Compare  $y_1$  &  $y_2 \Rightarrow = \Rightarrow$  invariant

$$a) y(n) = x(n/2)$$

$$\rightarrow y_1(n) = x\left(\frac{n}{2} - m\right)$$

$$\rightarrow y_2(n) = x\left[\frac{n-m}{2}\right]$$

$$y_1(n) \neq y_2(n) \quad \therefore \text{variant}$$

$$b) y(n) = x^2(n-2)$$

$$y_1(n) = x^2(n-2-m)$$

$$y_2(n) = x^2(n-m-2)$$

$$y_1(n) = y_2(n) \quad \therefore \text{invariant}$$



$$c) \quad y(n) = x(n) + nx(n-2)$$

$$y_1(n) = x(n-m) + nx(n-2-m)$$

$$y_2(n) = x(n-m) + n-m x(n-m-2)$$